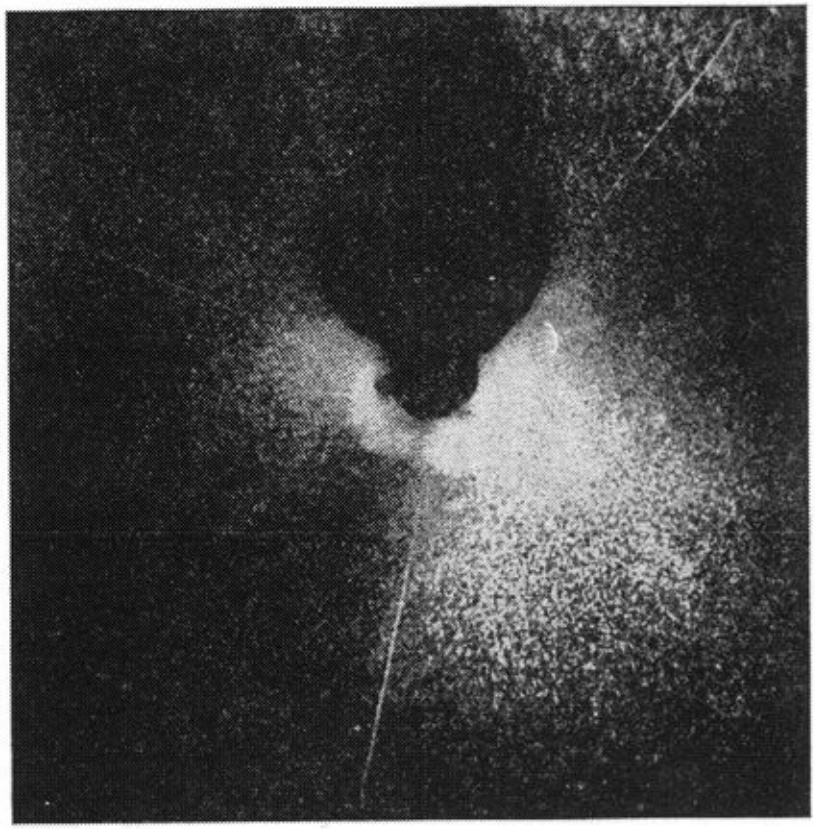


Figure 28.—Transmutation of a boron nucleus into three helium nuclei by a proton.

(Here, too, an alpha particle is visible, which does not happen to belong to this reaction.)

Finally, in view of the ancient history of these problems, let us ask whether we are able today to make gold out of mercury, which used to be the dream of the alchemists of old, and in what way this might be possible. To answer this question, it is sufficient to take a look at Table IV. It shows that mercury and gold are immediate neighbours, so that just one single step would be necessary to accomplish the transmutation of one into the other. In other words, by pure chance, the alchemists of past centuries happened to be on the right track when they endeavoured to turn mercury into gold. According to our present knowledge, mercury has 7 stable isotopes, with mass numbers ranging from 196 to 204, while gold has only one, with the mass number 197; this mass number is absent among the known isotopes of mercury. If the mercury isotope of mass number 196 is irradiated by neutrons, so that a neutron is incorporated in a mercury nucleus, the otherwise unknown nucleus of mass number 197 must be the result. This nucleus must be unstable, or else it would have been observed already. It will therefore change, by positron emission or *K*-radiation, into the stable gold nucleus of the same mass number. Thus the following two reactions occur, in this order:

(1) ${}_{80}\mathrm{Hg}^{196} + {}_{0}\mathrm{n}^{1} \rightarrow {}_{80}\mathrm{Hg}^{197}$